

CHAPTER 15

15.1) FOURIER ANALYSIS OF A PULSE

We complete the square in the exponent in (60) by writing

$$-Dk^2t + ikx = -(u-u_0)^2 + u_0^2 = -u^2 + 2uu_0 \quad , \quad (S1)$$

where $u \equiv k(Dt)^{\frac{1}{2}}$. Insertion and comparison of terms yields u_0 :

$$2uu_0 = 2k(Dt)^{\frac{1}{2}}u_0 = ikx \quad , \quad u_0 = ix/2(Dt)^{\frac{1}{2}} \quad .$$

If (S1) is inserted into (60) the integral may be written

$$\int_{-\infty}^{+\infty} dk \dots = \exp(u_0^2)(Dt)^{-\frac{1}{2}} \int_{-\infty}^{+\infty} \exp[-(u-u_0)^2] du \quad . \quad (S2)$$

The integral to the right is the Gauss integral -- see Appendix A -- with the integrand displaced by u_0 . This does not change the value of the integral, $\pi^{\frac{1}{2}}$. Therefore, the right side of (S2) has the value

$$(\pi/Dt)^{\frac{1}{2}} \exp(u_0^2) = (\pi/Dt)^{\frac{1}{2}} \exp(-x^2/4Dt) \quad .$$

Division by 2π , as in (60), yields (14).

15.2) DIFFUSION IN TWO AND THREE DIMENSIONS

(a) Because $r^2 = x^2+y^2$ or $r^2 = x^2+y^2+z^2$, the two forms (64) and (65) may be written as products of the form

$$\theta_2(t) = \theta(x,t)\theta(y,t) \quad , \quad \theta_3(t) = \theta(x,t)\theta(y,t)\theta(z,t) \quad , \quad (S1)$$

where

$$\theta(u,t) = c \times (4\pi Dt)^{-\frac{1}{2}} \exp(-u^2/4Dt) \quad , \quad u = x, y, z \quad , \quad (S2)$$

has the same form as (14), satisfying the one-dimensional diffusion equation

$$D \partial^2 \theta / \partial u^2 = \partial \theta / \partial t \quad . \quad (S3)$$

Insert now $\theta_2(t)$ into the two-dimensional form of (8):

$$\begin{aligned} & [D \partial^2 \theta(x,t) / \partial t^2] \theta(y,t) + \theta(x,t) [D \partial^2 \theta(y,t) / \partial t^2] \\ & = [\partial \theta(x,t) / \partial t] \theta(y,t) + \theta(x,t) [\partial \theta(y,t) / \partial t] \quad . \end{aligned}$$

Because both $\theta(x,t)$ and $\theta(y,t)$ satisfy (S3), the terms cancel out in pairs, and (8) is satisfied. The proof for three dimensions is analogous.

(b) From (S1) and (S2), and comparison with (64) and (65):

$$C_2 = c^2 / (4\pi D) \quad , \quad C_3 = c^3 / (4\pi D)^{3/2} \quad .$$

The two functions $\theta_2(t)$ and $\theta_3(t)$ will be normalized to unity when $c = 1$: In two dimensions, with the help of (15),

$$\iint \theta_2(t) dx dy = \left[\int_{-\infty}^{+\infty} \theta(x,t) dx \right] \left[\int_{-\infty}^{+\infty} \theta(y,t) dy \right] = 1 \quad .$$

The proof for three dimensions is again analogous.

15.3) TEMPERATURE VARIATIONS IN SOIL

The lowest temperature occurring in a temperature wave of the form (13), at a specified depth, but at an unspecified time, is

$$\theta_{\min} = -\theta_0 \exp(-z/\delta) \quad .$$

If both an annual and a daily wave are present, with the same amplitude $\theta_a = \theta_d$, but different frequencies ω_a and $\omega_d = 385.24 \omega_a$, superimposed on an average temperature θ_0 , the minimum temperature at depth z is

$$\theta_{\min} = \theta_0 - \theta_d [\exp(-z/L_a) + \exp(-z/L_d)] \quad ,$$

where $L_a = 1 \text{ m}$, $L_d = 1 \text{ m}/(365.24)^{\frac{1}{2}} \cong 0.0523 \text{ m}$. In our example, $\theta_{\min} = 0^\circ\text{C}$, $\theta_0 = \theta_d = 10^\circ\text{C}$. Hence we must solve the transcendental equation

$$\exp(-u) + \exp(-0.0523u) = 1 \quad , \quad (S1)$$

where $u = z/L_a$. Eq. (S1) is readily solved by iteration (see the solution to Problem 13.5), by writing it in the form

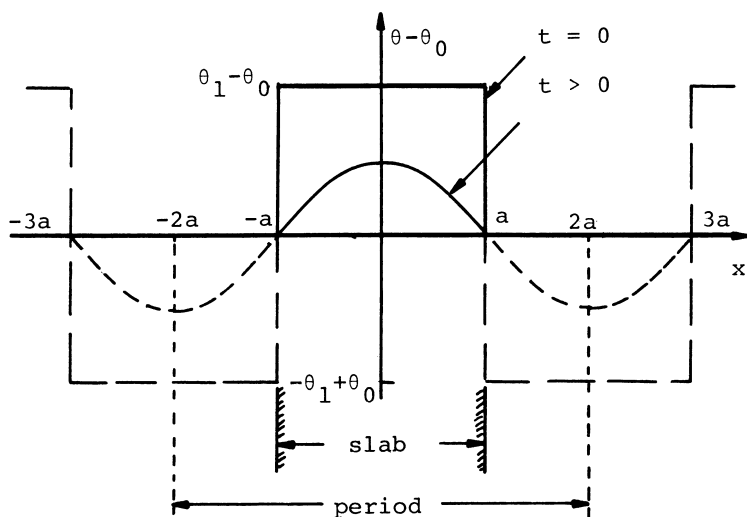
$$u = -0.0523 \log[1 - \exp(-u)] \quad .$$

Starting with the ad-hoc value $u = 1$, the iteration converges to about 0.116. Hence,

$$z \cong 12 \text{ cm} \quad .$$

15.4) COOLING OF A SLAB

We view the temperature in the slab as part of a spatially periodic temperature distribution with the period $4a$:



Such a periodic distribution can always be expanded into a Fourier series

$$\theta(x,t) - \theta_0 = \sum_{\substack{n \\ (\text{odd})}} A_n(t) \cos(n\pi x/2a) \quad , \quad (S1)$$

where the sum goes over all odd values of n . Every term in the series vanishes at $x = \pm a$. Thus, $\theta(x,t)$ satisfies the boundary condition $\theta(\pm a,t) = \theta_0$ at all times.

From the Fourier theorem, the coefficients A_n are given by

$$A_n(t) = \frac{1}{2a} \int_{-2a}^{+2a} [\theta(x,t) - \theta_0] \cos(n\pi x/2a) dx \quad .$$

Specifically, for $t = 0$:

$$A_n(0) = 4 \times \frac{2}{a} (\theta_1 - \theta_0) \int_0^a \cos(n\pi x/2a) dx = \frac{4}{n\pi} (\theta_1 - \theta_0) \quad , (S2)$$

where we have used the symmetry of the temperature distribution to reduce the integration interval.

Inserting (S1) into the heat conduction equation (8):

$$\sum_n \dot{A}_n \cos(n\pi x/2a) = -D \sum_n A_n (n\pi/2a)^2 \cos(n\pi x/2a) \quad .$$

In order for this to be satisfied for all values of x , this equation must hold for every value of n separately:

$$\dot{A}_n = -n^2 A_n / t_0 \quad , \quad 1/t_0 = D (\epsilon\pi/2a)^2 \quad .$$

This has the solutions

$$A_n(t) = A_n(0) \exp(-n^2 t/t_0) = \frac{4}{n\pi} (\theta_1 - \theta_0) \exp(-n^2 t/t_0) \quad ,$$

where we have used the initial conditions (S2). The term with $n = 1$ decays least rapidly with time; once $t \gg t_0$, all other terms will be negligibly small compared to it. The temperature at the center of the slab has fallen off to 0.01 of the initial temperature when

$$\theta_1(t) = (4/\pi)(\theta_1 - \theta_0) \exp(-t/t_0) = 0.01 \times (\theta_1 - \theta_0) \quad ,$$

$$t = t_0 \log(400/\pi) \cong 4.85t_0 \quad .$$

15.5) P-N JUNCTION:

DIFFUSION FROM A FIXED SURFACE CONCENTRATION

According to (23):

$$n_d(x) = n_d(0)[1 - \operatorname{erf}(z)] \quad , \quad z = x/(4Dt)^{\frac{1}{2}} \quad .$$

We have $n_d = n_a$ when

$$\operatorname{erf}(z) = 1 - n_a/n_d(0) = 0.9 \quad .$$

From tables of the error function, $\operatorname{erf}(z) = 0.9$ for $z = 1.16$. Hence, with $D = 10^{-13} \text{ cm}^2 \text{ s}^{-1}$:

$$x = (4Dt)^{\frac{1}{2}}z = (7.34 \times 10^{-7} \text{ cm s}^{-\frac{1}{2}}) \times t^{\frac{1}{2}} = Ct^{\frac{1}{2}} \quad .$$

15.6) HEAT DIFFUSION WITH INTERNAL SOURCES

In addition to (66) we still have (4). For purely radial heat flow in a cylinder or a sphere, (4) becomes

$$J_u = -K \, d\tau/dr \quad .$$

Rather than inserting this into (66) and solving the resulting equation (for $\partial/\partial t = 0$), we proceed in a more direct way: The total heat flowing outwards through the radius r must equal the total heat generated inside that radius. For a cylinder of length L with purely radial heat flow:

$$J_u \times 2\pi rL = -K \times 2\pi rL(d\tau/dr) = g_u \times \pi r^2 L \quad ,$$

$$d\tau/dr = -g_u r/2K \quad .$$

This must be true for any $r < R$, where R is the outer radius of the wire. Integration:

$$\theta(r) = -g_u r^2/4K + \text{constant} \quad ,$$

$$\tau(0) - \tau(R) = g_u R^2/4K \quad .$$

For a sphere:

$$J_u \times 4\pi r^2 = K \times 4\pi r^2(d\tau/dr) = g_u \times 4\pi r^3/3 \quad ,$$

$$d\tau/dr = -g_u r/3K \quad ,$$

$$\tau(r) = -g_u r^2/6K + \text{constant} \quad ,$$

$$\tau(0) - \tau(R) = -g_u R^2/6K \quad .$$

Note that in both cases the temperature rise is proportional to g_u/K and to the square of the linear dimension of the body taken in the direction of the heat flow. The only difference is in a shape-dependent numerical factor (1/4 vs. 1/6). This is a general rule for bodies of arbitrary shape.

15.7) CRITICAL SIZE OF NUCLEAR REACTOR

Eq. (3), augmented by a generation term n/t_0 , reads

$$\partial n/\partial t = n/t_0 - \text{div } \vec{J}_n \quad .$$

Insert $\vec{J}_n$ from (1):

$$\partial n/\partial t = n/t_0 + D_n \nabla^2 n \quad . \quad (S1)$$

The prospective solution (67) satisfies

$$\partial n / \partial t = n / t_1 \quad ,$$

$$\nabla^2 n = - (k_x^2 + k_y^2 + k_z^2) n \quad .$$

Insert into (S1) and cancel the common factor n :

$$1/t_1 = 1/t_0 - D_n \times (k_x^2 + k_y^2 + k_z^2) \quad .$$

With this value of t_1 , (67) is a solution of (S1). The reactor will be above-critical if $n(x,y,z,t)$ grows with time, that is, if $t_1 > 0$. This requires

$$k_x^2 + k_y^2 + k_z^2 < 1/D_n t_0 \quad . \quad (S2)$$

The k 's are not arbitrary. The boundary condition

$$n(x,y,z,t) = 0 \quad \text{for} \quad x,y,z = \pm L/2$$

requires that for each of the three k 's we have $\cos(kL/2) = 0$, or $k = n\pi/L$, where each of the three n 's is an odd integer. The smallest allowed values of k for which (S2) can be satisfied, are those with $n = 1$: $k_x = k_y = k_z = \pi/L$. Inserted into (S2) and re-arranged:

$$L^2 > 3\pi^2 D_n t_0 \equiv L_{\text{crit}}^2 \quad .$$